

Build the Judge Before the Strategy: A Reproducible Retail Backtesting Platform and a Five-Study Negative-Results Program on ASX/US Equities

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Abstract

We describe Project-Drift, a solo-researcher, reproducible backtesting platform for long-only equity strategies, and report the results of a five-study, six-candidate pre-registered evaluation program run on it against ASX and US equities. The platform’s central design commitment is that the evaluation machinery — a bitemporal point-in-time (PIT) data store, a structurally enforced features/labels leakage boundary, a walk-forward simulator with a costed and fairly constructed benchmark, and pre-registered Deflated Sharpe Ratio (DSR) and Probability-of-Backtest-Overfitting (PBO) gates — was built and frozen *before* any strategy return was computed. Four candidate strategies were evaluated against this fixed judge to a return-based verdict: (1) cross-sectional momentum with a trend overlay, (2) a franking-credit dividend-capture trade, (3) US insider cluster-buying, and (4) S&P/ASX 200 index-addition reconstitution. Two further candidates (LIC/LIT discount-to-NTA, post-earnings-announcement drift), reported together as a fifth study, were blocked at the data-feasibility stage by vendor data-access limits before any return could be computed. All six candidates resolve negative or blocked: no candidate clears its pre-registered promotion bar, and none is presently under active evaluation. We report the headline numbers for each study together with the concrete software and measurement bugs that pre-registration, mutation-tested gates, and adversarial review caught along the way, including a first-pass insider-cluster result of +871.5% mean return that a data-alignment bug had manufactured, which fell to a statistically indistinguishable −5.2% after three independent fixes ($N = 2,110$ paired events). The contribution of this paper is accordingly not the discovery of a profitable trading rule, but a demonstration that a pre-registered, reproducible evaluation framework can reject several plausible retail strategies while simultaneously detecting multiple implementation errors that would otherwise have produced false positives. We argue that five rejections from one disciplined evaluation harness are informative in aggregate, not merely five failures, and that the harness and the discipline, not any single strategy, are the durable output of this project.

This is a personal research project, not financial advice. Nothing in this paper is investment advice or a solicitation.

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1 Introduction

Retail systematic trading has a specific and well-known failure mode. A backtest is easy to build and easy to fool: look-ahead bias from vendor-adjusted prices, survivorship bias from delisted names quietly dropping out of a data feed, and outright overfitting from tuning parameters against the same window used to judge them will all, independently and in combination, manufacture an attractive equity curve. Because the failure is silent (the backtest itself reports no error, only a number), it typically surfaces only much later, as live losses that the backtest never predicted. Unlike a mis-specified regression, a backtest that lies to its own author gives no diagnostic; it just looks good.

The response taken in this project is an inversion of the usual build order. Rather than building a strategy and then trying to convince oneself the backtest is trustworthy, the evaluation machinery is built first, hardened against the standard ways a backtest can be fooled, and its statistical promotion criteria are pre-registered before any strategy is run against it. Concretely: a bitemporal point-in-time (PIT) data store that can only be queried “as of” a historical decision date; a structural boundary, enforced by both a static test and a linter rule, that makes it impossible for feature-construction code to import forward-looking labels; a walk-forward simulator whose benchmark is exactly as costed and point-in-time-constrained as the strategy it is compared against; and a Deflated Sharpe Ratio / Probability-of-Backtest-Overfitting gate pair whose thresholds, sweep grid, and trial-counting mechanism were frozen by decision record before the first real-data backtest ran [Bailey and López de Prado, 2014, Bailey et al., 2017].

Four candidate strategies were run to a return-based verdict against this fixed judge (evaluation window 2026-06-23 – 2026-07-11): a classical cross-sectional-momentum-plus-trend book, a franking-credit dividend-capture trade, a US insider cluster-buying signal, and an S&P/ASX 200 index-addition reconstitution trade. Two further candidates, evaluated together as a fifth, data-feasibility study, were blocked by data-vendor access limits before a return could even be computed. **All six candidates resolve negative or blocked.** No candidate clears its pre-registered promotion bar; the platform currently has no strategy under active evaluation.

Contributions. This paper offers three things. First, a description of the platform design as actually implemented (the PIT store, the leakage boundary, the cost/benchmark model, and the statistical gates), verified directly against the source code rather than against its docstrings (§3). Second, five pre-registered negative (or blocked) results reported with the same template (hypothesis, pre-registered rule, method, result, verdict) so they can be compared and audited uniformly (§5). Third, a catalogue of concrete overstatement failure modes that were actually caught in the course of this work, not hypothesized in the abstract (§6). One instance: the first pass of the insider cluster-buying study reported a mean six-month return of +871.5%, driven by a price-basis mismatch between an unadjusted entry price and a post-split-adjusted exit price. After three independent bugs were found and fixed by adversarial review, the corrected, otherwise-identical result was a statistically indistinguishable −5.2%. A second, much smaller instance ran in the opposite direction: a pre-existing unit bug that fed annualized rather than per-period Sharpe ratios into the trial registry had been silently crushing every historical DSR value toward zero; correcting it moved one strategy’s DSR from $\approx 2.4 \times 10^{-8}$ to 0.40 (still short of the pre-registered bar, but well above the uncorrected value). Both corrections were caught by process (pre-registration discipline, mutation-tested gates, and a second adversarial reviewer), not by luck, and both are reported here alongside the studies they affected rather than silently absorbed into a single “final” number.

Motivation from a prior project. The discipline described in this paper did not emerge in the abstract. In a separate, earlier project of the author’s (a machine-learning-ranked equity strategy, not part of the Project-Drift repository), a backtest reported an eye-catching $\approx 42\%$ CAGR. A forensic autopsy of that project found the number to be an artifact of four compounding errors: the model had effectively been tuned against its own out-of-sample Sharpe ratio, a normalization step had been fit on the whole history (including the test window), the reported configuration was a best-of-many selection with no trial-count correction, and the backtest was not survivorship-clean. That experience is what motivated building the evaluator first and freezing its promotion criteria before running any strategy through it, rather than a hypothetical concern. To be clear about scope: this repository’s own `MLRanker` component is, and has always been, an unimplemented stub that raises `NotImplementedError` and has never been run; no result, number, or study from the separate prior project appears anywhere in this paper’s results or tables (§5); it is described here in prose only, as motivating background for the platform design in §3. This paper reports the five studies that were actually run on Project-Drift itself.

To state the thesis plainly before the evidence: this paper’s contribution is not a profitable trading rule (none of the six candidates evaluated here is one) but a demonstration that a pre-registered, reproducible evaluation framework can reject several plausible retail strategies while simultaneously catching multiple implementation errors that would otherwise have produced false positives. The remainder of the paper supports that claim: §3 describes the framework, §5 reports the five rejections it produced, and §6 catalogues the errors it caught along the way.

2 Related work

The statistical apparatus used to judge every strategy in this paper is not novel; it is adopted directly from the backtest-overfitting literature. The Deflated Sharpe Ratio (DSR) corrects a strategy’s Sharpe ratio for the number of independent trials evaluated, non-normality of returns (via skewness and kurtosis), and the length of the track record, producing a one-sided test of whether an observed Sharpe ratio is significantly positive after accounting for selection bias [Bailey and López de Prado, 2014]. Sharpe ratio estimation itself is not asymptotically as simple as annualizing a monthly figure by $\sqrt{12}$; the correct treatment of serial correlation and the proper variance of an estimated Sharpe ratio is due to Lo [2002], whose skew/kurtosis-adjusted variance this project’s DSR implementation uses directly. The Probability of Backtest Overfitting (PBO), estimated via combinatorially symmetric cross-validation (CSCV), asks a complementary question: across all ways of splitting a track record into an in-sample and out-of-sample half, how often does the configuration that looked best in-sample turn out to be below-median out-of-sample [Bailey et al., 2017]. Both statistics, together with the broader pre-registration and purged/embargoed cross-validation discipline this project follows, are treated at length in López de Prado [2018].

The momentum and trend-following strategy evaluated in §5.1 descends from the classical cross-sectional momentum anomaly documented by Jegadeesh and Titman [1993] and the time-series (“trend-following”) variant documented by Moskowitz et al. [2012]. The franking/dividend-capture study in §5.2 is a variant of the classical ex-dividend price-drop-ratio literature; Elton and Gruber [1970] is the foundational reference relating the ex-dividend price drop to investors’ marginal tax rates and to dividend-clientele effects, which is the same tax-asymmetry mechanism (here, Australian franking credits) this project’s spike attempts to exploit. The insider cluster-buying study in §5.3 uses a delisting-return convention adapted from Shumway [1997], who documents that naively treating delisted names as flat or missing (rather than assigning a realistic terminal loss) biases small-cap backtests upward: exactly the survivorship-style distortion this project’s platform

is built to avoid. The index-reconstitution study in §5.4 sits inside a well-documented literature on the index-inclusion price effect; [Chen et al. \[2004\]](#) is an early asymmetric treatment of S&P 500 additions and deletions, and [Greenwood and Sammon \[2025\]](#) documents that the abnormal return to US index inclusion, historically large, has fallen from an average of roughly 7.4% in the 1990s to 0.3% in the most recent decade for additions (and to 0.1% for deletions over 2010–2020), a secular decline consistent in direction, though not directly comparable in market or magnitude, with this project’s null result on ASX 200 additions.

3 Platform design (methods)

This section describes what the platform’s code actually does, verified against the source (file and line references in `FACTS.md` §A–D), not against docstrings or aspirational comments.

3.1 The point-in-time bitemporal store

The core data primitive is a store keyed on two independent date axes per row: an *effective date* (the trading day a fact is about) and a *knowledge date* (the day that fact became knowable). A query “as of” some historical date D filters every row to `knowledge_date` $\leq D$, then keeps the latest-known version of each (`symbol`, `effective_date`) pair under a deterministic tie-break order. Prices are stored raw; adjusted close and adjusted volume are computed only at query time from the raw price series plus a separate, dated corporate-actions table. The store itself never persists a vendor-supplied adjusted price. A set of in-memory, LRU, and universe-membership performance caches, plus a crash-safe file-compaction operation, sit on top of this raw, as-of query path and are proven, by test, to be entirely result-neutral; their mechanics are described in [Appendix B](#).

3.2 The features/labels leakage boundary

The platform’s feature-construction code (`features/build.py`) is structurally prevented from ever seeing a forward-looking label. Feature construction filters its input panel to `effective_date` $<$ `as_of` before computing anything (the code comment on this line calls it out explicitly as “the Mutation point”), while label construction (`labels/build.py`) is, deliberately and by contrast, forward-looking: it queries prices as of the *end* of the forward-return horizon, and its own docstring states plainly that it “must only be joined to features inside training/evaluation code, never in the live path”. This boundary is enforced twice, independently, by a static test and an independent lint rule, so that a change to one enforcement mechanism alone cannot silently open the leak; the enforcement mechanics are described in [Appendix B](#). A related, distinct test proves that a later-arriving correction to the store (written with a future knowledge date) cannot silently change the byte-identical result of a historical as-of read.

The platform also ships a purged-and-embargoed cross-validation splitter (`backtest/cv.py`), with a one-month label-horizon purge and a twelve-month embargo pre-registered under ADR-0012. Because the one signal actually deployed to date, a classical cross-sectional momentum model, does no model fitting, this splitter is currently dead code for the shipped path: real, unit-tested, and present for the future ML phase, but not exercised by any of the five studies reported here. We flag this explicitly rather than imply the purge/embargo machinery has been load-bearing in any result reported in this paper.

3.3 Corporate-action reconstruction and the adjusted-close hazard

The platform’s data provider for EOD Historical Data (EODHD) ingests only raw OHLCV plus dated `/splits` and `/div` feeds, and never reads the vendor’s `adjusted_close` field; adjusted-close reads are guarded against by a runtime assertion, whose exact trigger condition is given in Appendix B. The motivating hazard for this design is concrete and measured directly in this repository’s earlier free-data spike: BHP’s 2013 Adj Close, as reported by a free vendor (Yahoo Finance via `yfinance`), was only 44% of its raw 2013 close, a 55.9% divergence, because that vendor’s adjusted series retroactively bakes in dividends paid, and a demerger executed, *after* 2013. Any momentum feature computed on that adjusted series “as of 2013” would silently embed information from 2015. Dividend adjustment inside the platform’s own pipeline uses the standard proportional factor $f = (P - D)/P$, where P is the raw close on the last bar strictly before the ex-date. Demergers, which have no vendor-supplied adjustment factor at all, are handled by a researcher-maintained overlay that is applied *only if* the raw price series shows an ex-date drop at least half the size of the distribution, a data-driven gate added after the unconditional version of the same overlay was found, on real data, to have injected a phantom +7.6% (BHP) / +26% (Woolworths) one-day jump where none should exist. On the real BHP and Woolworths demerger dates, the vendor’s raw series is in fact already continuous, so the gate correctly makes the overlay a no-op for both seed events.

A separate identity-resolution layer treats ISIN, not ticker code, as the primary key for a security, because ASX ticker codes are reused: the code FGL, for example, resolves today to an unrelated post-2011 penny stock rather than the historical Foster’s Group (delisted 2011). Any ambiguity in this resolution fails closed: the resolver refuses to merge two records rather than guess.

3.4 Walk-forward engine, cost model, and the fair benchmark

The walk-forward simulator rebalances on a configurable cadence (monthly by default) and enforces a strict decision/fill separation: a decision made at date D may only use data with knowledge date $\leq D$ and effective date $< D$, and the corresponding trade fills a fixed number of bars later (one, by default) rather than at the decision bar’s own close. Trading costs are a flat 15 basis points per trade (5 bps brokerage plus a 10 bps half-spread), with an additional slippage term of 5 bps at baseline that scales with the trade’s participation in the name’s trailing average daily dollar volume, tripling under an optional stress mode. A flat-rate capital-gains-tax haircut model exists in code but is currently configured to a rate of exactly zero, so it is a functioning no-op in every result reported in this paper: the platform has not yet modeled taxes or franking credits as a return drag in any live number.

The benchmark every strategy is compared against is not a passive index but a costed, point-in-time, periodically-reconstituted equal-weight portfolio of the *same* universe the strategy itself sees at each decision date, run through the identical cost model and the identical broker simulator. This design explicitly rejects what an internal decision record calls “the costless true-buy-and-hold escape hatch”: an earlier version of the benchmark had been built as-of the *end* of the backtest window, survivor-biased, forward-adjusted, costless, and drawn from a broader universe than the strategy’s own point-in-time universe. That bug was found and fixed before any of the results in this paper were computed. A sanity check on synthetic random-walk data confirms the benchmark is not rigged to lose: over 2018–2020 synthetic data the benchmark’s Sharpe ratio (≈ 3.1) legitimately exceeds the strategy’s (≈ 1.2).

3.5 DSR, PBO, and the machine-binding gates

Every reported result is passed through two pre-registered statistics. The Deflated Sharpe Ratio is computed with Lo’s skew/kurtosis-adjusted variance of the Sharpe-ratio estimator [Lo, 2002] and compared against an expected-maximum-Sharpe benchmark computed via the López de Prado selection-bias correction [Bailey and López de Prado, 2014], using the Euler–Mascheroni constant $\gamma \approx 0.5772156649$. The pre-registered significance bar is $\text{DSR} \geq 0.95$, a one-sided $p < 0.05$ test that the deflated Sharpe ratio is significantly positive. The Probability of Backtest Overfitting is computed via combinatorially symmetric cross-validation with $S = 16$ contiguous partitions of the return series, giving $\binom{16}{8} = 12,870$ in-sample/out-of-sample splits, with a pre-registered bar of $\text{PBO} < 0.20$ [Bailey et al., 2017]. The frozen, finite momentum-lookback sweep evaluated for the PBO calculation is $\{6, 8, 9, 10, 11, 12, 13, 14\}$ months, eight configurations, with lookback 12 pre-registered as the primary deliverable.

Two further mechanisms exist specifically to prevent the gate from being gamed after the fact: the trial count N fed into the DSR’s selection-bias correction carries a committed, non-resettable floor (currently 8, equal to the sweep size), and a *binding* gate mode refuses to emit a PASS off a reset or never-swept trial registry. Both the gate logic and the trial-counting invariant are covered by mutation-style unit tests: tests that assert the gate actually flips to FAIL under a broken guard, not merely that it passes when the guard is intact. The floor computation and the mutation-test mechanics are given in Appendix B.

3.6 A concrete unit bug in the DSR pipeline

One defect is worth describing in the methods section rather than only in the results, because it affected the DSR gate’s correctness across every historical run, not just one study. The trial registry’s contract requires per-period (i.e., monthly, unannualized) Sharpe ratios; the walk-forward engine had instead been registering the *annualized* Sharpe ratio into that registry. Because the DSR calculation’s expected-maximum-Sharpe term scales with the variance of registered Sharpe values, this silently inflated that term by a factor of roughly $\sqrt{12} \approx 3.46$, crushing every historical DSR toward zero regardless of the strategy’s true skill. The bug was caught by an external model-based review, fixed in the engine, and the historical registry was migrated (each historical Sharpe value divided by $\sqrt{12}$) with explicit researcher approval before the migration was applied. For the primary momentum configuration evaluated in §5.1, the corrected DSR moved from $\approx 2.4 \times 10^{-8}$ to 0.40, still well short of the 0.95 bar but well above the uncorrected value. Because the study’s mechanical pre-registered verdict depended only on the Sharpe confidence interval, not on DSR, this correction changed the substantive read of the result without changing its mechanical verdict. Consistent with the project’s one-shot-window discipline (§9), the other seven configurations in the same sweep were left uncorrected rather than re-run, and are flagged as unreliable pending a freshly-scoped future evaluation.

4 Data

4.1 Commercial data: strengths and a documented gap

The platform’s paid data source is EOD Historical Data (EODHD), an EOD-only subscription tier (no fundamentals/earnings-calendar entitlement) with a daily rate limit of 100,000 API calls. This vendor is, on the evidence gathered in this project, a reasonably deep and inexpensive source of US and AU end-of-day history, but it has one specific, material, documented gap: the project’s history-depth gate (requiring 80% coverage of a sampled delisted population reaching back to a

target decade) failed on real data at a coverage of 0.7857. Investigation traced this not to an artifact of sampling irrelevant micro-caps, but to a genuine coverage hole concentrated in large, liquid ASX delistings: of thirteen well-known ASX delistings from 2005–2015 with deal sizes ranging from roughly A\$1.3 billion to US\$14 billion (including WMC Resources, Rinker, AXA Asia Pacific, Patrick Corporation, and David Jones), twelve were either completely absent from the vendor’s feed or reachable only under a ticker code since reused by an unrelated company. A subsequent attempt to recover these twelve names by cross-referencing ISIN codes and alternate symbols against the vendor’s full 4,373-row symbol table recovered zero of twelve cleanly. Coverage from 2017 onward, by contrast, is strong (mostly 100% per year against an independent 69-company delisting census), giving a cumulative coverage of $36/41 = 0.878$ against the 0.80 bar. An external statistical review noted, however, that the 95% Wilson confidence interval on that point estimate, $[0.745, 0.947]$, dips below the bar, so even the “passing” window is a fragile pass, not a guaranteed one. This tradeoff, a 2017-present window with usable coverage, at the cost of excluding the 2008–09 financial crisis and the August 2011 selloff, and at a statistical-power cost of roughly 30–45% versus a full-history window at equal DSR confidence, is the one used for the momentum study in §5.1 and the index-reconstitution study in §5.4.

Separately from any strategy evaluation, a one-time bulk archive of the EODHD subscription was run to completion (equity `eod` coverage: 97.36% for AU, 91.63% for US), specifically so the paid subscription could eventually be cancelled while the already-purchased history is retained permanently. The resulting coverage report currently reports `safe_to_cancel = false` for both markets: 95 AU and 2,678 US delisted equity symbols are still missing price history, almost entirely attributable to a known EODHD vendor limitation (a pre-listing zero-price-placeholder artifact, not a bug in the ingestion code) rather than a genuine coverage gap. The same vendor that has a real large-cap ASX delisting gap also provides deep, cheap coverage of the post-2017 period and of US equities generally.

4.2 Free/public data used elsewhere

Two of the five studies in this paper use data sources outside the paid EODHD feed. The insider cluster-buying study (§5.3) uses the US Securities and Exchange Commission’s free, structured Form 3/4/5 quarterly datasets; a provenance check on 5,447 open-market purchase filings confirmed the reported filing date lags the underlying transaction date (median lag 2 days, 0% of filings dated before the transaction they report), ruling out a naive same-day-knowledge assumption. The index-reconstitution study (§5.4) uses a hand-built dataset assembled from S&P Dow Jones Indices’ own public quarterly-review press releases, which are not licensed vendor data. Every row in that dataset was manually transcribed from parsed PDF text after an automated summarization tool was found, on at least one occasion, to fabricate entirely fictitious index changes that were not present in the source document, a caution about tool reliability, not about the underlying public source, that is disclosed here for transparency.

5 Results: five pre-registered studies

Each study below follows the same template: hypothesis, pre-registered rule and threshold (with a cryptographic hash where a frozen dataset was involved), method, result, and verdict. All numbers are drawn from `FACTS.md`; the letter-and-number codes (e.g.) point to the corresponding row.

5.1 Study 1: Cross-sectional momentum with a trend overlay

Hypothesis. A classical cross-sectional momentum ranking, combined with a time-series trend overlay, earns a risk-adjusted return in excess of a fair, costed, point-in-time ASX equity benchmark [Jegadeesh and Titman, 1993, Moskowitz et al., 2012].

Pre-registered rule. Anchored to a specific git commit before any 2017-present data was ingested: PROCEED requires the net-Sharpe 95% confidence-interval lower bound > 0.30 , at least half of the eight swept configurations to clear $DSR \geq 0.95$, and the primary configuration’s PBO < 0.20 ; STOP requires the confidence-interval upper bound ≤ 0.00 ; anything else is INCONCLUSIVE.

Method. The full 2017-01-01 to 2026-07-02 window was ingested for 3,709 candidate ASX symbols (active and delisted common stock), and the walk-forward engine run at monthly cadence for 115 periods.

Result. The primary (12-month lookback) configuration’s net Sharpe ratio was 0.62 with a 95% confidence interval of $[-0.02, 1.27]$; gross Sharpe 0.74, net CAGR 6.3%, net Sortino 0.57, net max drawdown -18.4% , monthly turnover 36%. The fair, costed, point-in-time benchmark’s Sharpe over the identical window was 0.645, nominally *above* the strategy’s own net Sharpe. The PBO across the eight-configuration sweep was 0.347, failing the pre-registered < 0.20 bar. The primary configuration’s DSR, after the unit-bug correction described in §3.6, was 0.40, failing the 0.95 bar. Three real-data defects (a missing core dependency, an alphabetic universe truncation that had silently excluded large blue-chip names, and a crash on one corrupted vendor dividend record) were found and fixed before any performance number existed; a residual gap of 10 of 3,709 symbols (0.27%) that still failed to ingest was disclosed and deliberately left unpatched, per the project’s one-shot-window discipline.

Verdict. Mechanically **INCONCLUSIVE**: the confidence interval clears neither bound. Substantively read as **no edge**: net Sharpe is statistically indistinguishable from (and nominally below) the fair benchmark, and PBO fails its bar. The strategy was formally parked, not deployed, and the vendor upgrade (Norgate) needed to resolve the pre-2017 coverage gap and obtain a full-history, crisis-inclusive verdict was explicitly not purchased to chase this result.

5.2 Study 2: Franking-credit dividend capture

Hypothesis. Australian franking (dividend imputation) credits create an exploitable, tax-driven ex-dividend price-drop asymmetry of the kind documented for the pre-imputation US market by Elton and Gruber [1970], tradeable via a short cum-to-ex round trip.

Pre-registered rule. GO required a credit-excluded (45-day-rule-honest), market-adjusted, net-of-cost effect $\geq +0.15\%$, a Welch t -statistic ≥ 2.0 , and a combined sample of at least 40 events across two groups of at least 15 each.

Method. A pre-registered, frozen universe of 12 large-cap “franker” names (banks, major miners, major industrials) was compared against 8 unfranked or foreign-earner control names (REITs and foreign-earner miners) over 2017-01-01 to the present, using only raw close plus ex-dividend-date data already on the researcher’s EOD-only subscription tier. The primary metric deliberately excludes the franking credit itself, since claiming it legally requires a 45-day at-risk holding period the short round trip does not satisfy; the credit-inclusive number is reported only as an explicitly unreachable ceiling.

Result. 233 franker events and 140 control events were pulled with zero data-quality events skipped. The franker mean return was -0.664% ; the control mean was $+0.237\%$; the effect (franker minus control) was -0.901% with a Welch t -statistic of -5.81 on a combined $N = 373$. All twelve individual franker names showed a negative mean return. Even the explicitly unreachable credit-

inclusive ceiling (+0.457% franker mean) exceeds the control by only +0.22%.

Verdict. STOP. The effect is not merely below the GO threshold; it is the wrong sign, and the significance is on the wrong side of zero as well. This was the one study of the five with no headline-number-changing implementation bug found in review. A narrower, 45-day-held variant was surfaced by an adversarial reviewer as a possible future idea but was explicitly not started and would need its own independent pre-registration.

5.3 Study 3: US insider cluster-buying

Hypothesis. Clusters of open-market purchases by multiple company insiders within a short window predict positive abnormal forward returns.

Pre-registered rule. GO required an annualized, net-of-cost, market-adjusted effect of at least +4.0% over a matched random control, a Newey-West heteroskedasticity-and-autocorrelation-consistent t -statistic of at least 2.0, and a combined sample of at least 40 events across two groups of at least 15 each.

Method. A cluster was defined as open-market purchases by at least two distinct officers or directors of the same issuer within a rolling 10-trading-day window, drawn from free SEC EDGAR Form 3/4/5 quarterly filings after the vendor’s own insider-transactions feed proved access-gated. The universe was restricted to a market-cap proxy under \$500 million, price above \$1, and average daily dollar volume above \$50,000. Entry was the first trading day’s open after the *filing* date (never the transaction date), held a fixed 126 trading days, subject to a fillability cap and roughly 600 basis points of round-trip cost at that cap, with delisted names assigned a Shumway-style terminal loss convention [Shumway, 1997] rather than dropped. The primary comparison was against a random matched control (matched on liquidity decile, market-cap decile, and industry, fixed seed). Of 93,696 qualifying purchase filings, 2,531 cluster events and 3,379 single-insider events survived the pre-registered floors, with 2,110 of the cluster events successfully paired to a matched control.

Result: first pass. The initial run reported a cluster mean six-month return of +871.5% against a control mean of +791.1%, an annualized paired-difference effect of +160.9%, and a Newey-West t -statistic of 0.16. **Result: corrected.** An independent adversarial review, conducted before the result was finalized, found and fixed three bugs: (i) entry prices were computed on raw, unadjusted opens while exit prices used the vendor’s split-adjusted close, so a post-exit stock split retroactively rescaled only one leg of the trade, manufacturing spurious multi-thousand-times returns on affected names (e.g. one ticker’s apparent return moved from \$2.82 to an “exit” of \$11,100); (ii) the Newey-West statistic’s input series was ordered by an internal grouping key rather than by calendar date, as the pre-registration required; and (iii) a single degenerate estimated market beta (−2,428, against a normal range of roughly −2 to +3.5 across the other 4,219 events) was producing a mathematically impossible market-adjusted return, fixed with a symmetric ± 5.0 beta clip. After all three fixes, on the identical $N = 2,110$ paired sample, the cluster mean return was −5.22% and the control mean was −5.28%; the paired-difference effect was +0.066%, an annualized +0.131%, with a Newey-West t -statistic of 0.021. A secondary, non-gating diagnostic found roughly two-thirds of *all* names in this micro-cap universe lose money net of costs over six months regardless of insider activity.

Verdict. STOP, decisively, both before and after the correction: the first-pass result already failed on significance alone ($t = 0.16 \ll 2.0$), and the corrected result fails on both magnitude and significance (+0.13% $\ll$ +4.0%; $t = 0.02 \approx 0$). A second, independent adversarial pass confirmed the entry timing used the filing date (never the transaction date), that the beta-estimation window never overlapped the holding period, and that the delisting convention did not flatter losing positions. That same review flagged one provenance caveat worth disclosing directly: the spike’s

code was uncommitted (untracked in git) at the time the corrected run executed, so the run is anchored to a git SHA for the surrounding repository state, not to a committed snapshot of the spike code itself. One further incident is disclosed for completeness: an uncaught exception during data collection briefly embedded an API credential in a local diagnostic log; the log was deleted, a repository-wide search confirmed no other copy existed, and the triggering code path was hardened. No dedicated architecture decision record exists for this study’s STOP verdict, unlike the other four candidates in this paper; the dated research-log entry is its only formal record, which we note as a minor documentation-discipline gap.

5.4 Study 4: S&P/ASX 200 index-addition reconstitution

Hypothesis. Stocks added to the S&P/ASX 200 index earn a positive announcement-to-effective-date abnormal return, consistent with the historical (if, per [Greenwood and Sammon \[2025\]](#), weakening) index-inclusion effect documented for US indices by [Chen et al. \[2004\]](#).

Pre-registered rule and data freeze. A dataset of S&P/ASX 200 and 300 constituent changes was hand-built from S&P DJI’s own public quarterly-review press releases: 55 of 58 possible quarters (March 2012 – June 2026) were captured, yielding 390 total addition-and-deletion events, of which 234 fall in the coverage-verified 2017-present window with 98.3% of those price-covered. After excluding two confirmed, disclosed demerger-driven fast-track additions that fall outside the normal quarterly review cycle (Coles Group and Endeavour Group), the primary, long-only-additions population was frozen and cryptographically hashed *before any return was computed*: 114 addition events across 30 independent quarters (2017-06 to 2026-06), SHA-256 50098b13...98a1b11. The pre-registered method buys at the open one trading day after the announcement, exits at the open on the announced effective date, market-adjusts against a point-in-time reconstituted equal-weight top-100 benchmark, applies the platform’s standard cost model, clusters by calendar quarter for significance, and registers as exactly one statistical trial (PBO was pre-registered as not computable, since no legitimate parameter grid exists for a single frozen population). PROMOTE-TO-PAPER required all three of: mean $\geq +1.00\%$ per event, 95% confidence-interval lower bound $> 0\%$, and DSR ≥ 0.95 . The pre-registration explicitly flagged, in advance, that 30 independent clusters is a low-power sample, comparable to the already-documented low-power problem in Study 1.

Result. A first run reported DSR = 0.034; independent adversarial review found that trading costs had been computed on the market-adjusted excess return rather than the stock’s own traded return, an error independently quantified at no more than 1.3 basis points per event against a roughly 43-basis-point mean cost drag and a 385-basis-point confidence-interval width, immaterial to the verdict, but corrected and re-run rather than left in place. Both runs are disclosed in the trial registry (28 total trials, not 27) rather than the first being silently discarded. The final, corrected result: quarter-clustered mean net-of-cost, market-adjusted return of -0.40% per quarter, 95% confidence interval $[-2.33\%, +1.52\%]$; event-level mean -0.34% , median -0.21% , 48.2% of events positive; DSR = 0.021 against the 0.95 bar. A non-gating deletions diagnostic showed a mean return of -0.07% , also indistinguishable from zero.

Verdict. STOP. None of the three pre-registered promotion conditions is met: the point estimate is negative (not merely below the $+1.00\%$ floor), the confidence interval does not clear zero, and the DSR is roughly 45 times below its bar. This closes the index-reconstitution candidate; as of this result, no strategy candidate is under active evaluation on the platform. This null result is independently corroborated by top-journal evidence that the US index-inclusion effect itself has decayed from roughly 7.4% in the 1990s to 0.3% (additions) / 0.1% (deletions, 2010–2020) in the most recent decade [[Greenwood and Sammon, 2025](#)]. The ASX 200 finding in this paper is consistent with a vanishing effect rather than an underpowered test of a still-large one.

5.5 Study 5: Two candidates blocked on data access

Two further candidates identified alongside the index-reconstitution study, a discount-to-net-tangible-assets signal on ASX-listed investment companies (LICs/LITs), and a post-earnings-announcement drift (PEAD) signal, were triaged for data feasibility on the researcher’s real, EOD-only EODHD subscription tier before any pre-registration was attempted. Both were blocked outright. The vendor’s fundamentals endpoint, which would be required for per-fund NTA/NAV history, returned HTTP 403 (not entitled) on every probe attempted, and no endpoint modeling closed-end-fund NTA or NAV exists anywhere in the vendor’s API surface at any subscription tier, a genuine product gap rather than an entitlement one. A local, network-free liquidity diagnostic on 20 known ASX LICs found zero of the twenty below a \$50,000/day fillability floor, so liquidity was not the binding constraint. The earnings-calendar endpoint needed for PEAD similarly returned HTTP 403 (not entitled) on both bulk and per-symbol pulls. Both candidates were closed as **BLOCKED**: unlike Studies 1–4, no return-based backtest was ever run for either, because no usable data existed to run one against.

5.6 Summary table

Study	Verdict	N	Headline effect	Significance / gate
1. Momentum + trend	PARKED (INCON- CLUSIVE / no edge)	115 monthly periods, 8-config sweep	net Sharpe 0.62 vs. benchmark 0.645	95% CI $[-0.02, 1.27]$; PBO 0.347 (<0.20 bar); DSR 0.40 (≥ 0.95 bar)
2. Franking dividend capture	STOP	373 (233+140)	-0.901%	$t = -5.81$ (bar: $t \geq 2.0$, effect $\geq +0.15\%$)
3. Insider cluster buying	STOP	2,110 paired events	$+0.131\%$ annualized (naive first pass: $+871.5\%$ mean 6-month level; naive annualized diff $+160.9\%$)	NW $t = 0.021$ (bar: $t \geq 2.0$, effect $\geq +4.0\%$; naive $t = 0.16$)
4. S&P/ASX 200 additions	STOP	114 events / 30 quarters	-0.40% /quarter	95% CI $[-2.33\%, +1.52\%]$; DSR 0.021 (≥ 0.95 bar)
5. LIC/NTA + PEAD	BLOCKED	N/A	N/A (no return computed)	N/A: HTTP 403 on all relevant endpoints

Table 1: All five studies resolve negative or blocked. See `FACTS.md §K` for the sourced version of this table.

6 A taxonomy of backtest-overstatement failure modes

Across the five studies of this evaluation program, the same small set of failure modes recurred in different guises. We catalogue them here, each tied to the concrete study where it was caught, because the catalogue generalizes better than any individual result does.

Adjusted-price look-ahead.

A vendor’s “adjusted” price series is computed as of *today*, so any historical bar of it silently

encodes every corporate action that happened between that bar and today. Measured directly: a free-data BHP series' 2013 adjusted close was 44% of its raw 2013 close, a 55.9% divergence, from baking in 2015 corporate actions. The platform's entire design, raw-price ingestion with a runtime guard against ever reading an adjusted field, exists specifically to close this hole.

Survivorship bias via vendor coverage gaps.

Even a paid, professional-grade vendor can have a coverage hole that silently removes exactly the large, real losses a strategy should have suffered. Twelve of thirteen well-known large ASX delistings from 2005–2015 were missing or contaminated in this project's paid feed, which is why the platform's history-depth gate exists as a machine check rather than an assumption.

Best-of-N without deflation.

Evaluating several parameter configurations and reporting only the best one systematically overstates the true edge. The platform's response is the DSR/PBO gate pair with a committed, non-resettable trial-counter floor, and the concrete value of this discipline is visible in Study 1: the PBO of 0.347 on an eight-configuration sweep would have been invisible to a single-run Sharpe report.

Price-basis mismatch (mixed adjustment bases).

The largest single distortion in this paper: computing an entry price on one adjustment basis (raw) and an exit price on another (split-adjusted) manufactures returns of several thousand percent on any name that split between the two dates. This drove Study 3's first-pass +871.5% mean result, corrected to -5.22% once both legs were placed on the same basis.

Non-chronological significance testing.

A Newey-West heteroskedasticity-and-autocorrelation-consistent test is only valid if its input series is ordered by the time dimension the autocorrelation structure assumes. Study 3's series was inadvertently ordered by an internal grouping key instead of by entry date, understating the correct standard error until corrected.

Unclipped market-adjustment inputs.

A single degenerate estimated beta, from a short trailing-window regression on a thin small-cap name, can produce a market-adjusted return below -100%, a mathematical impossibility that nonetheless silently entered a downstream mean. Both Study 3 and the platform's broader fill-math hardening (untradeable-bar handling for zero/NaN/garbage prices) exist to close variants of this hole.

Cost basis computed on the wrong leg.

Study 4's first-run DSR of 0.034 rested on a cost calculation applied to the market-adjusted excess return rather than the stock's own traded return; the resulting distortion was small (independently bounded at ≤ 1.3 basis points per event) but was found, quantified, fixed, and disclosed rather than left in a "good enough" first-pass number.

Sharpe-ratio unit errors.

Annualizing (or failing to un-annualize) a Sharpe ratio before feeding it into a statistic whose scale assumption depends on the period length is an easy, quiet error. This project's own DSR pipeline had exactly this bug for an unknown period of its history, systematically

making every reported DSR look worse than it truly was (§3.6). It is the one failure mode in this catalogue that understated, rather than overstated, an edge.

Missing or unfair benchmark.

A costless, forward-adjusted, survivor-biased, or broader-universe benchmark makes any strategy look better by comparison. The platform’s benchmark redesign under an internal decision record exists specifically because an earlier version of this exact bug was found in this project’s own code before Study 1 was run.

Data-access illusions.

Not every candidate fails on results; two failed before a result could even be computed, because the required data simply is not obtainable on the researcher’s subscription tier (§5.5). Discovering this *before* spending engineering effort on a backtest that could never be run is itself a discipline, not a failure to report.

7 Discussion

Five pre-registered evaluations, all negative or blocked, is not bad luck; under a properly calibrated statistical gate it is close to the expected base rate for a solo researcher screening plausible-sounding retail strategies without a structural information edge. The alternative explanation, that the gate itself is too strict, or that a sixth or seventh candidate would surely clear it, is exactly the reasoning that a disciplined pre-registration protocol is designed to make expensive to act on impulsively: any change to the DSR or PBO threshold, the sweep grid, or the trial-counting mechanism requires a new, dated decision record, not a quiet code edit.

The value of the discipline is not abstract; it is visible in the specific bugs it caught. Pre-registration before data collection meant Study 4’s cost-basis bug and Study 3’s price-basis mismatch were caught and disclosed rather than silently baked into a single reported number. Mutation-tested gates meant the DSR/PBO thresholds could not silently degrade into a no-op. A committed, non-resettable trial-counter floor meant the momentum study’s selection-bias correction could not be quietly reset by deleting a local file. A second, adversarial reviewer, sometimes an external model, sometimes an internal subagent explicitly tasked with skepticism, caught issues a single reviewer plausibly would have missed, including both the annualized-Sharpe unit bug (§3.6) and all three bugs behind Study 3’s spurious result. This pattern, process catching bugs that careful individual authorship did not, generalizes beyond the specific strategies tested.

Limitations. This is a single-researcher project evaluated over roughly three weeks; the sample of five studies is itself small, and we do not claim the taxonomy in §6 is exhaustive. The retail data tier used throughout has real depth limits: the ASX momentum study is confined to a 2017-present window with disclosed statistical-power costs relative to a full-history evaluation, and two candidates could not be evaluated at all for want of a fundamentals-tier data feed (§5.5). Several of the five studies’ pre-registered clustering units are themselves small, 30 independent quarters for Study 4, disclosed in advance as comparable to Study 1’s already-flagged low-power breadth problem, so an INCONCLUSIVE or narrowly-STOP verdict in this paper should be read as “no edge demonstrated at the power available,” not as definitive proof that no edge could exist under any specification. Finally, even carefully written, unit-tested code needed a second, adversarial pass to be trustworthy in this project. The implementation was not written carelessly; the review layer, not raw authorial care, is treated here as the operative safeguard precisely because of that.

8 Conclusion

Five candidate strategies were evaluated against a single, fixed, pre-registered evaluation harness over a three-week period; all five resolved negative or were blocked before a result could be computed. We consider this a successful, not a wasted, research program. The platform, the bitemporal PIT store, the structurally enforced leakage boundary, the fairly costed benchmark, and the mutation-tested DSR/PBO gates, is retained as strategy-agnostic infrastructure and is, in our view, the durable output of this work; the parked and stopped strategy code is retained in-tree as tested reference implementations rather than deleted. No strategy candidate is presently under active evaluation. The decision to stop searching and simply index, rather than continue testing successively weaker candidates against the same harness, is itself a decision the harness’s own discipline recommends.

9 Reproducibility statement

Every real-data run reported in this paper is anchored to a specific git commit SHA, a SHA-256 hash of the parameter configuration file, and, where a frozen dataset was involved, a SHA-256 hash of that dataset. For example, the Study 4 primary population’s hash, 50098b134f7d58caa7058217db657ada786b761947b7. Every logged experiment follows a fixed template recording the data range, universe, parameters, config hash, git SHA, and run artifact directory. A one-shot-window discipline governs what may be fixed after a result exists: defects found and fixed *before* any performance number was computed are in scope; anything found *after* is disclosed but deliberately left unpatched within that window, so as not to take a second, informed look at data that has already produced a result. An internal adversarial review pattern, a second reviewer or subagent explicitly tasked with skepticism, layered on top of an initial correctness pass, caught the cost-basis bug in Study 4 and confirmed the bug fixes in Study 3; external model-based review caught the DSR unit bug and the fragility of the Study 1 coverage window. Mutation-style unit tests, deliberately breaking a guard condition and confirming the test suite catches it, not merely that it passes when the guard is intact, back the leakage boundary, the DSR/PBO gates, and the corporate-action adjustment logic. Four automated gates are required green before any commit under this project’s own working conventions: strict-mode type checking, lint (including the banned-import leakage rule), formatting, and the non-integration unit test suite.

All code, configuration, and decision records referenced in this paper are in the **Project-Drift** repository. The paper itself, including this reproducibility statement, was assembled by tracing every quantitative claim back to a specific file and line in that repository, recorded in `paper/FACTS.md`; no number in this paper is not sourced there.

References

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A Appendix: sourced facts table (human-readable summary)

The complete, sourced facts table underlying every claim in this paper is maintained as `paper/FACTS.md` in the repository (approximately 140 individually cited rows, organized into sections A–M). We reproduce the five-study summary and the two headline before/after corrections here for convenience; see `FACTS.md` for the full table with file-and-line citations for every number.

Correction	Before	After
Study 3 cluster mean 6-month return	+871.5% (naive)	−5.22% (corrected, $N=2,110$)
Study 3 Newey-West t -statistic	0.16	0.021
Study 4 primary-configuration DSR	0.034 (trial 27)	0.021 (trial 28)
Study 1 primary-configuration DSR	$\approx 2.4 \times 10^{-8}$	0.40 (unit-bug fix; still fails 0.95 bar)

Table 2: Headline before/after corrections across the studies.

B Appendix: implementation notes

This appendix collects the low-level implementation mechanics that the body of the paper (§3) refers to by a one-line pointer. Each paragraph below backs a specific claim already stated in the body; nothing here changes a result, and every number is the same one cited in `FACTS.md`.

Performance caching and compaction (§3.1). Three caching layers sit on top of the raw, as-of query path, and are proven not to change results: an in-memory materialisation of each store directory, parsed once per run via a single `DuckDB read_parquet` call; an as-of-keyed LRU query cache; and a PIT-consistent universe-membership cache. A separate `compact()` operation coalesces small parquet files into fewer large ones without deduplicating or reordering the underlying row multiset, and only deletes the source files after the coalesced file has been fully written, so a crash mid-compaction cannot lose data.

Leakage-boundary enforcement (§3.2). The features/labels boundary is enforced twice, independently: a static test walks the abstract syntax tree of every file under `features/` and asserts that none of them imports anything from `project_drift.labels`, and a ruff lint rule (`flake8-tidy-imports` banned-API) independently bans the same import at lint time, with a message string that names the leakage risk directly.

Adjusted-close runtime guard (§3.3). A runtime assertion raises an error if any ingested column normalizes to `adjustedclose` or `adjclose`, so an accidental future code path that reads a vendor-adjusted field fails loudly at ingestion time rather than silently poisoning the PIT store.

Trial-registry floor and mutation tests (§3.5). The global count of statistical trials N fed into the DSR’s selection-bias correction is read from a persisted, append-only registry, and a committed *floor* value (currently 8, equal to the sweep size) guarantees that $N = \max(\text{floor}, \text{live registry count})$, so deleting the local, gitignored registry file cannot reset the effective trial count below 8. A *binding* gate mode additionally requires the *live* registry count (not the floored one) to actually meet the floor before it will emit a PASS, refusing to bind a positive verdict off a reset or never-swept registry. Both the gate logic and the trial-counting invariant are covered by mutation-style unit

tests that assert the gate actually flips to FAIL when DSR or PBO is pushed below threshold, and that a reset registry raises rather than silently defaulting to $N = 1$.

C Appendix: pre-registration anchors

- **Study 1** (momentum + trend, 2017+ window): git SHA f1136eae190f3a49a929fd02dbf3cb83a7b88dcd, pre-registered 2026-07-02.
- **Study 2** (franking): pre-registered 2026-07-04, git SHA 9a9d7a40821dbf63b075474b1b955cfce325d11f (per docs/research-log.md STEP-0 entry).
- **Study 3** (insider cluster): pre-registered 2026-07-05, git SHA 3c92a7ea989df7d0955d0237842d170720d4cb1 (STEP 0/1); final corrected run anchored to git SHA 6c4cde55b6c4c598d0ee3069383634d2d87bbc1c (per docs/research-log.md).
- **Study 4** (index additions): git SHA 6c4cde55b6c4c598d0ee3069383634d2d87bbc1c, configs/base.yaml SHA-256 87e1dfc93686a1a3adfb73c70154e80283f66ddc1b76719326f9ffa69f784443, frozen event-set SHA-256 50098b134f7d58caa7058217db657ada786b761947b70b090355efa1b98a1b11, timestamp 2026-07-11T10:46:12Z.
- **Study 5** (LIC/NTA, PEAD): data-feasibility triage dated 2026-07-10, no pre-registration reached (blocked before that stage).